Natural Language Processing (NLP)

- NLP is a branch of artificial intelligence that focuses on the interaction between computers and humans through natural language
- **The goal** of NLP is to enable machines to understand, interpret and respond to human language. For example:
    - Understand: NLP systems parse customer feedback to identify key themes and sentiments
    - Interpret: Determine the intent behind queries in virtual assistants
    - Respond: Chatbots generate relevant responses to user inquiries in real-time
- **Applications:**
    - Text Classifications: Identifies themes like spam or sentiment within texts
    - Machine Translation: Converts text or speech across languages
    - Speech Recognition: Transforms spoken words into text
    - Chatbots and Virtual Assistants: Facilitate interactive dialogues, mimic human conversation
    - Information Extraction: Detects specific data like names or relationships from texts
    - Text Summarization: Condenses long documents into essential summaries
- **Challenges:**
    - Ambiguity: Language is inherently ambiguous
    - Context Understanding: NLP struggles with context
    - Sarcasm and Irony: Detecting sarcasm or irony in text is challenging
    - Language Diversity: Different dialects and slang complicate processing
    - Idioms and Phrases
    - Pronoun Resolution
    - Language Variability and Evolution: Language is constantly evolving, and new words, slang, and usage patterns emerge all the time.
- **History:**
    - Early Days (1940s - 1960s): Early models were based on rules of grammar and the formal theory of syntax
    - Rise of ML (1970s - 1990s): There was a shift towards statistical methods for language processing, influenced by the limitations of rule-based systems
    - The Internet Era (1990s - 2000s): The explosion of digital text data on the internet provided vast resources for training and improving NLP systems. Significant advances in machine translation during this period, driven by statistical models
    - DL and LLM Era (2010s - present):
        - DNNs like CNNs, RNNs, LSTMs significantly improved NLP and outperformed earlier methods in many language tasks
        - Pre-trained Language Models: BERT and GPT used large-scale pre-training to learn rich language representations, greatly improving language understanding and text generation

    - LLMs are scaled-up pre-trained models, often with more parameters, larger training data, reasoning ability and multimodal capabilities
    - Agentic AI: LLM-based systems can plan, use tools and take actions, extending NLP towards broader intelligent systems

Key Concepts

1. Part-of-Speech Tagging (POS Tagging)

The process of labelling each word in a sentence with its word type, such as noun, verb, adjective, … based on its meaning and context

2. Named Entity Recognition (NER)

A process where an algorithm takes a string of text (sentence or paragraph) and identifies relevant nouns (people, places and organizations) that are mentioned in that string

3. Syntax Tree

A tree representation of syntactic structure of sentences or strings

4. Semantic Analysis

An area of NLP focused on interpreting the meaning of words, sentences and texts in context

5. Word Embeddings

A type of word representation that allows words with similar meaning to have a similar representation

6. N-grams

- A sequence of n words, characters or tokens in text or speech
  Examples:
  Unigram (1-gram): [“the”], [“cat”], [“sleep”]
  Bigram (2-gram): [“the cat”], [“the dog”]
  Trigram (3-gram): [“The cat sleeps”], [“The girl moans”]
- Not all N-grams are kept. We usually filter them by frequency, keeping only sequences that appear frequently in the corpus
- By analysing frequently occurring sequences in a corpus, N-grams can capture language patterns and provide more meaningful information than single words alone

7. Word Cloud

A word cloud visually represents text data, highlighting frequently used words in varying sizes and / or color based on their occurrence

8. Term Frequency

- **TF (Term Frequency)** measures how frequently a term occurs in a document. More occurrences increase the term's relevance in that document

  $$TF = \frac{Number\ of\ times\ term\ appears\ in\ a\ document}{Total\ number\ of\ terms\ in\ the\ document}$$

- **TF-IDF (Term Frequency - Inverse Document Frequency)** weighs a TF against its rarity across all documents, enhancing its importance when it is rare

  $$IDF = ln(\frac{Total\ number\ of\ documents}{Number\ of\ documents\ with\ the\ term\ in\ it})$$

  $$TF - IDF = TF \times IDF$$

- IDF tells us how important a word is by looking at how rare it is across all documents

9. DTM / TDM

- **Document Term Matrix (DTM):** This is a matrix where each row represents a document and each column represents a term (word) from the corpus. The entries in the matrix typically contain the frequency of the term in the document
- **Term Document Matrix (TDM):** This matrix is the transpose of the DTM, each row represents a term and each column represents a document. Similarly, the entries indicate the frequency of the term in the corresponding document